INSTITUT FOR MATEMATIK
AARHUS UNIVERSITET

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Statistical Models
Week 14: Wrapping up

This is the last teaching week of the semester; we finished the last topic of Bayesian statistics Friday, 28th november.

The three classes in course week 14 are:

- Tuesday: Exercises 13,
- Wednesday: Analysing a real data set, a workshop. We hope this will be interesting and fun!
- Friday: last lecture. I will recap the course together with you and we have a look at the exam topics.

Relevant Readings

Lecture Material

- Link to Google drive

Data Workshop on Wednesday

Data: An Agricultural Experiment About Ants

A researcher interested in ants and their benefit for agriculture in the Midwest, Karl A. Roeder, has posted a dataset on USDA, https://doi.org/10.15482/USDA.ADC/28454771.
The data set contains two .csv files: Metadata and Master Data, you can download them as `Ant_Experiment.csv` and `Metadata for Ant Tillage Experiment.csv` from the Google Drive.
The Metadata describes column variables for the Master Data file that contains all of the information used in the analyses. This project encompasses two years of work (12 weeks per year) where ants were sampled along 8 transects of corn with tillage, 8 transects of corn without tillage, 8 transects of soybeans with tillage, and 8 transects of soybeans without tillage. Sentinel prey stations were deployed for 48hrs after ant sampling and contained ~60 corn earworm eggs to provide a quantitative assessment of pest predation by beneficial ants.

Tentative methods from the paper
(= background information from authors)

Study Site
Research was conducted at the Eastern South Dakota Soil and Water Research Farm in Brookings, South Dakota, USA (44.352882, -96.803453). Historically, multiple crops have been grown at this location including corn (Zea mays) and soybean (Glycine max) with and without tillage (hereafter referred to as Till and No-Till). In 2015, four ~0.4ha blocks were setup in a complete block design that each included four 50m x 20m equal sized plots of till corn, no-till corn, till soybean, and no-till soybean. The crops in each of these plots were rotated yearly with the other crop (corn to soybean and soybean to corn), but the tillage treatment of spring disking was kept in place such that the same plots were always tilled or not tilled.

Ant activity: abundance and species richness

Ants were sampled weekly in all plots from June to August in both 2021 and 2022. We setup two equidistant transects within each plot (N=32 transects per week) that were spaced at least 20m apart. Within each transect, we evenly placed on the ground twenty 1.5ml conical microcentrifuge tubes (Midland Scientific, USA) that contained a cotton ball that was saturated with a 20% sucrose solution. This baiting technique is common for ant studies measuring activity and has been used in numerous studies across a variety of ecosystems. For all sample weeks, tubes were deployed at 0800, collected after one hour, and returned to the lab for processing (i.e. counting the number of ants and species) and data analysis. The morning sampling time was chosen based on pilot trials in 2020 that indicated ants would not forage earlier do to potentially cold temperatures in the morning (e.g. <10°C) or during the middle of the day when temperatures were too hot (e.g. >50°C). These limiting abiotic conditions were exacerbated in June when vegetation was sparse.
Environmental variables (air, ground, and soil temperatures along with relative humidity) were measured in between bait tube deployment/pickup at the beginning, middle, and end of each transect during each sampling event. Air temperature and relative humidity were measured using a Kestrel 3000 weather station (Kestrel Instruments, USA), ground temperature was measured using a Fluke 62 max IR thermometer (Fluke Corporation, USA), and soil temperature was measured using a Weber instant-read digital meat thermometer (Weber, USA). Values from each variable were then averaged per transect for further analysis.

Measuring pest predation using sentinel prey

We measured predation of pest insects as one ecosystem service provided by ants during each sampling period (N=24 total weeks) by placing one sentinel prey station in the crop row directly north of each transect immediately after bait tubes were collected. Each sentinel prey station was comprised of ~60 freeze-killed corn earworm eggs (Helicoverpa zea; Frontier Agricultural Sciences, USA) that were glued to a small piece of a flash card and then placed in a small petri dish within a vertebrate exclosure. Corn earworm eggs have regularly been used as sentinel prey for ants in agricultural studies due to their relevance as a crop pest. After 48hrs in the field, sentinel prey were collected and the remaining eggs were counted to quantify the proportion of pest insects that had been attacked.

from the same author(s): https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecs2.3645